

A Quasiperiodic Product of Countably Many Orthogonal Projections Need Not Converge in Norm

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Status. Candidate counterexample, likely valid, for human review. It gives a negative answer to the countably infinite quasiperiodic-family question recorded in arXiv:1809.05858.

1 Source question and answer

Let (M_j) be closed subspaces of a Hilbert space, let P_j be their orthogonal projections, and set

$$x_n = P_{\sigma(n)}x_{n-1}.$$

For a finite family, Sakai proved norm convergence when σ is *quasiperiodic*: every index occurs infinitely often and the gaps between successive occurrences of each fixed index are bounded.

On printed page 16 of Ginat's dissertation [1], the following question is attributed to Sakai:

Does norm convergence still hold when the family is countably infinite and the index sequence is quasiperiodic?

Theorem 1. *There are a countable family $(M_j)_{j \geq 1}$ of closed subspaces of ℓ_2 , a quasiperiodic map $\sigma : \mathbb{N} \rightarrow \mathbb{N}$, and $x_0 \in \ell_2$ such that the sequence*

$$x_n = P_{\sigma(n)} \cdots P_{\sigma(1)}x_0$$

does not converge in norm. In fact, $x_n \rightharpoonup 0$, $\lim_n \|x_n\| > 0$, and $\bigcap_j M_j = \{0\}$.

2 Idea of the proof

Move a unit vector from e_m to e_{m+1} in very many tiny planar rotations. The sum of the squared angles is finite, so projecting through the hyperplane implementing each rotation loses only a positive finite factor. Nevertheless, the unit directions escape to infinity and converge weakly to zero.

Each hyperplane is initially used exactly when its rotation is needed. It is then withheld until the path has left the two coordinates supporting its normal vector. After that time it may be repeated forever without changing the orbit. Exact 2-adic valuation classes provide a disjoint bounded-gap recurrence schedule for all the old hyperplanes simultaneously. This makes the complete index sequence quasiperiodic while preserving the one-pass nonconvergent orbit exactly.

3 The slowly rotating path

Let $(e_m)_{m \geq 1}$ be the canonical orthonormal basis of $\mathcal{H} = \ell_2$. Put

$$N_m = 2^m, \quad \theta_m = \frac{\pi}{2N_m}.$$

Starting at $v_0 = e_1$, join e_m to e_{m+1} by the N_m unit vectors at successive angles $\theta_m, 2\theta_m, \dots, N_m\theta_m$:

$$\cos(r\theta_m)e_m + \sin(r\theta_m)e_{m+1}, \quad 1 \leq r \leq N_m.$$

Concatenating these finite strings defines $(v_k)_{k \geq 0}$. Every v_k is a unit vector, and consecutive vectors lying in stage m satisfy

$$\langle v_{k-1}, v_k \rangle = \cos \theta_m =: c_k > 0.$$

Moreover $v_k \rightharpoonup 0$, since the two coordinates supporting v_k tend to infinity. At the end of stage m the path equals e_{m+1} , so it does not converge in norm.

For every $k \geq 1$, define

$$u_k = \frac{v_{k-1} - c_k v_k}{\sqrt{1 - c_k^2}}, \quad Q_k = I - u_k \otimes u_k.$$

Thus u_k is a unit vector orthogonal to v_k , Q_k is the orthogonal projection onto $u_k^\perp$, and

$$Q_k v_{k-1} = c_k v_k. \tag{1}$$

If k belongs to stage m , then $u_k \in [e_m, e_{m+1}]$. Consequently there is an integer $L_k \geq k$ (for example, the end of stage $m+1$) such that

$$v_j \perp u_k \quad \text{for every } j \geq L_k. \tag{2}$$

Let $a_0 = 1$ and $a_k = \prod_{i=1}^k c_i$. The infinite product has a strictly positive limit. Indeed,

$$\sum_{k \geq 1} (1 - c_k) = \sum_{m \geq 1} N_m (1 - \cos \theta_m) \leq \frac{1}{2} \sum_{m \geq 1} N_m \theta_m^2 = \frac{\pi^2}{8} \sum_{m \geq 1} \frac{1}{N_m} < \infty.$$

Since every $c_k > 0$, this gives

$$a_k \longrightarrow a_\infty > 0. \tag{3}$$

4 A bounded-gap recurrence schedule

Write $\nu_2(n)$ for the exact exponent of 2 dividing the positive integer n . Let

$$P_1 = I, \quad P_{k+1} = Q_k \quad (k \geq 1),$$

and define $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ as follows:

1. $\sigma(2k-1) = k+1$ for $k \geq 1$;
2. if n is even, $\nu_2(n) = k+1$ for some $k \geq 1$, and $n \geq 2L_k$, set $\sigma(n) = k+1$;
3. at every remaining even integer, set $\sigma(n) = 1$.

This sequence is quasiperiodic. Index 1 occurs at least at every integer $n \equiv 2 \pmod{4}$, so its gaps are at most 4. For fixed $k \geq 1$, index $k+1$ first occurs at $2k-1$. After a finite delay it occurs at every sufficiently large integer satisfying $\nu_2(n) = k+1$. Those integers form an arithmetic progression with common difference 2^{k+2} . Hence the gap between successive occurrences of $k+1$ is bounded (the one initial gap is finite as well).

5 Exact orbit and failure of norm convergence

We claim that for every $k \geq 1$,

$$x_{2k-1} = x_{2k} = a_k v_k. \quad (4)$$

This follows by induction. At odd time $2k-1$, the prescribed projection is Q_k , so (1) sends $a_{k-1}v_{k-1}$ to $a_k v_k$. At even time $2k$, either the projection is I , or it is some Q_i whose recurrence has been released. In the latter case $2k \geq 2L_i$, hence $k \geq L_i$, and (2) gives $Q_i v_k = v_k$. Thus the even step does nothing, proving (4).

Equations (3) and (4) show that $\|x_n\| \rightarrow a_\infty > 0$, while $v_k \rightarrow 0$ gives $x_n \rightarrow 0$. Therefore (x_n) cannot converge in norm.

Finally, the common intersection is zero. During stage m there are at least two distinct normal vectors u_k in $[e_m, e_{m+1}]$, and they span that two-dimensional coordinate plane. A vector belonging to every $u_k^\perp$ is therefore orthogonal to e_m and e_{m+1} for every m , so it is zero. This proves Theorem 1.

6 Bounded literature audit

The source itself records the countably infinite quasiperiodic case as open. A later paper of Eskandari and Moslehian [2] proves convergence in a different *pseudo-periodic* regime: only a finite recurrent core is used, and its countable tail is subject to additional structure. That result does not apply here, because every one of the countably many hyperplane indices is recurrent with bounded gaps. Exact-title and exact-phrase searches of arXiv and the run indexes found no later source containing the counterexample above. This is a bounded novelty check, not a claim of exhaustive priority.

References

- [1] O. Ginat, *The Method of Alternating Projections*, arXiv:1809.05858 (2018), especially Section 3.4.3, printed p. 16.
- [2] R. Eskandari and M. S. Moslehian, *Convergence of Random Products of Countably Infinitely Many Projections*, arXiv:2405.04848 (2024).